

Project 21

Reachable Workspace of Flexiv Enlight L with Fixed q_3

Format: Group of two students; 2 ECTS; simulation-based project.

Prerequisites: URDF modelling, inverse kinematics, Jacobians, singularities, and numerical sampling.

Motivation

Fixing the third joint of Flexiv Enlight L produces a **6R robot parameterized by $\lambda = q_3$** , with up to eight regular shoulder–elbow–wrist IK branches. Joint limits and singularity margins can restrict their reachable regions differently. A workspace picture without branch labels can hide these differences.

Task

Overall objective. Compare the reachable position workspaces of all eight branches at several fixed q_3 values. Hold flange orientation fixed within each map: these are orientation-conditioned workspaces, not the complete six-dimensional pose workspace.

- **Construct and validate the 6R family.** Identify the base, joint axes, physical limits, and flange frame in the supplied model. Let $u = (q_1, q_2, q_4, q_5, q_6, q_7)$ and define $F_\lambda(u) = F(q_1, q_2, \lambda, q_4, q_5, q_6, q_7)$. Keep λ fixed within each map. Compare reduced and original URDF forward kinematics at 20 configurations per parameter value. Remove column 3 from the full geometric Jacobian to obtain the 6×6 reduced Jacobian; check it by finite differences.
- **Enumerate and identify the eight branches.** Use the geometric construction below to implement the positioning-arm and spherical-wrist IK. Use persistent three-sign labels, retain all eight slots when solutions are absent, and verify full-pose closure. Test three diagnostic poses and shuffled solver output. Near a separator, flag ambiguity or singularity instead of forcing a sign. A general-purpose 7R solver is not required.
- **Build comparable workspace maps.** At $\lambda = 30^\circ$, choose two distinct fixed flange orientations from validated configurations and record their numerical rotation matrices. Evaluate the IK branches on a common three-dimensional position grid. For each branch, distinguish absent real IK, excessive pose residual, joint-limit violation, and insufficient regularity. Produce eight aligned panels and a map counting how many branches are feasible at each position. Show empty branch workspaces explicitly.
- **Measure the effect of fixing q_3 .** Repeat the study at $\lambda = -30^\circ$ and 60° for the first orientation only, using the same base frame, position grid, and tolerances. Compare each branch's occupied volume, the union of all branches, and their pairwise overlap. Explain differences using geometry, joint limits, and the branch-specific failure masks; do not assume every branch or parameter value must give a distinct workspace.
- **Validate a decisive boundary.** Refine one boundary or narrow region. Check at least 20 inside/near-boundary targets independently using numerical IK or local continuation with q_3 locked. Validate any illustrated connection at intermediate poses, checking continuity, limits, and reduced regularity. Neighboring feasible voxels alone do not establish a feasible motion.

Specifics and scope

Use exactly three q_3 values, two orientations for the baseline and one shared orientation for the other two values: four workspace cases in total. Begin with approximately 30^3 voxel centers per case in a documented box covering the model’s reach. Refine one case or a selected subregion to half the voxel spacing. Estimate volume as the number of feasible voxels times voxel volume; report resolution sensitivity rather than claiming exact volume. Use identical grids for overlap comparisons. A convex hull must not be used as a substitute for the reachable set because it can fill holes and gaps.

Use the joint-position limits of the supplied URDF consistently; distinguish them from any separate safety limits. Resolve equivalent angles modulo 2π without discarding physically admissible representatives, and retain continuous representatives when checking motion. Divide the translational Jacobian rows by a stated characteristic length $\ell > 0$ before comparing singular values. Record the position, orientation, and singularity tolerances and vary the regularity threshold in the selected refinement study.

Here, the eight requested “aspects” are operationally identified by the eight fixed- q_3 shoulder–elbow–wrist branches. Distinguish these labels from connected components of the limit-feasible set; filtering can disconnect a branch. A singularity of the reduced 6×6 Jacobian need not be a singularity of the full 6×7 Jacobian, which has no determinant. The core study is kinematic: meshes support geometric interpretation, but collision-free, dynamic, and hardware feasibility are outside scope.

Geometric starting point

The robot description (.urdf) and link meshes (.stl) will be provided to the students. Implement the analysis in a robotics library or your own code; no additional solver or starter dataset is assumed. Verify the following dimensions and frame conventions against the supplied model before using them. For the flange pose (p, R) , define the shoulder center S , wrist-center vector w , and arm lengths in metres:

$$S = (0, 0, 0.260)^T, \quad a = 0.290, \quad b = 0.280, \quad w = p - 0.182Re_z - S, \quad e_z = (0, 0, 1)^T.$$

At fixed $\lambda = q_3$, the two elbow choices and two shoulder choices follow from

$$\begin{aligned} c_4 &= \frac{\|w\|^2 - a^2 - b^2}{2ab}, & q_4 &= \pm \arccos(c_4), \\ v &= (b \sin q_4 \cos \lambda, b \sin q_4 \sin \lambda, a + b \cos q_4)^T, & x &= \pm \sqrt{w_x^2 + w_y^2 - v_y^2}, \\ q_1 &= \text{atan2}(w_y, w_x) - \text{atan2}(v_y, x), \\ q_2 &= \text{atan2}(w_z, x) - \text{atan2}(v_z, v_x). \end{aligned}$$

Reject non-real cases, allowing only a declared round-off tolerance. Recover the two wrist flips from the residual orientation, accounting for the fixed URDF frame rotations. Label a regular solution by $(\text{sgn } x, \text{sgn } \sin q_4, \text{sgn } \sin q_6)$. Check the reduced determinant’s proportionality to $abx \sin q_4 \sin q_6$ under your Jacobian convention. The shoulder separator is $x = 0$, not simply $q_2 = 0$.

Expected outcome

Deliver reproducible code, the selected parameters and orientations, a branch-labelled workspace atlas, volume and overlap tables, and a refinement and boundary-validation example. Submit a concise 5–6-page report and a short simulation demonstration. Explain which conclusions concern the fixed- q_3 6R family and which cannot be extended to the unrestricted seven-joint robot.